(Please write your Enrollment Number) Enrollment No. 194

MID-TERM EXAMINATION

B. Tech. CSE-AI Semester: 6th

(March 2026) OFFLINE mode

Subject Code: BAI 322	Subject: Data Warehousing and Business Intelligence
Time: 1 ½Hours	Maximum Marks: 30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
a)	Differentiate between an independent data mart and a dependent data mart.	2.5	CO1
b)	Identify two common data anomalies that must be handled during the Data cleaning phase of pre-processing.	2.5	CO1
c)	Define a 'Fact Table' and a 'Dimension Table' within a Multi-Dimensional Data Model.	2.5	CO2
d)	Define a 'hyper cube' in the context of OLAP in the Data Warehouse.	2.5	CO2
	UNIT I		
Q2	**Attempt any two parts**	(5*2=10)	
a)	Explain the Three tier architecture of a data warehouse and provide an overview of its components.	5	CO1
b)	Discuss the ETL Process in detail. Why is Data transformation often considered the most complex and time-consuming step in this pipeline?	5	CO1
c)	Explain the lifecycle of defining business requirements in DW. How do information packages support dimensional design?	5	CO1
	UNIT II		
Q3	**Attempt any two parts**	(5*2=10)	
a)	An e-commerce company records sales transactions using the dimensions Time, Product, Customer, and Location. Design a suitable Star Schema for this scenario by identifying the fact table and dimension tables (assume appropriate attributes where necessary).	5	CO2
b)	Explain the slice-and-dice and pivot (rotation) operations with the help of a suitable example and diagrams.	5	CO2
c)	Explain the Fact Constellation Schema with a suitable example. Show how multiple fact tables share common dimension tables.	5	CO2

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